

Figure 7.20: Colony morphology of *Mycobacterium tuberculosis*on LJ medium

HOTS

M. tuberculosis the world's most deadly pathogen why?

7.11.5 Laboratory Diagnosis

Specimen: In case of pulmonary tuberculosis the most usual specimen is sputum.

Direct Microscopy: Smear is made from the sputum specimen and stained by Ziehl – Neelson technique. It is examined under oil immersion objective lens. The acid fast bacilli appear as bright red bacilli against a blue background.

Culture: The specimen is inoculated onto LJ – medium and incubated at 37°C for 2 weeks the tubercle bacilli usually grow in 2–8 weeks. The bacterial growth is confirmed by Ziehl – Neelsonstaining.

1. Tuberculin Skin test

Mantoux test: This method has been used routinely. In this test 0.1 ml of PPD (Purified protein derivative) containing 5 TU (Tuberculin unit) is injected intradermally on the flexor aspect of forearm (Figure 7.21) The site is examined after 48–72 hours and induration are measured (diameter in mm).

Positive test: Indurations of diameter d10 mm or more is considered positive.

Negative test: Indurations of 5 mm or less is negative.

2. Gene Xpert MTB

It is an automated diagnosis test it detects DNA sequences specific for *M. tuberculosis* and rifampicin resistance by PCR. Results can be obtained within 2 hours.

7.11.6 Treatment

The antitubercular drugs include two types of agents which are;



Figure 7.21: Mantoux test

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